Number Theory Tutorial

Modular Arithmetic and Chinese Remainder Theorem

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Problems

Problem 1.

Compute the following:

(a) $137 \pmod{12}$,

(b) $2^{10} \pmod{7}$,

(c) $3^8 \pmod{10}$,

(d) $17^5 + 23^4 \pmod{6}$.

Problem 2.

Find the remainder when

$$7^{100}$$

is divided by 13.

Problem 3.

Solve the following congruences:

(a)
$$7x \equiv 5 \pmod{12};$$

(b)
$$15x \equiv 9 \pmod{21};$$

(c)
$$18x \equiv 12 \pmod{30}.$$

Give all solutions modulo the corresponding modulus.

Problem 4.

Find the multiplicative inverse of each number modulo the given modulus:

(a) $7^{-1} \pmod{26}$,

(b) $11^{-1} \pmod{30}$,

(c) $37^{-1} \pmod{101}$.

Problem 5.

Solve the system

$$\begin{cases} x \equiv 3 \pmod{7}, \\ x \equiv 5 \pmod{11}. \end{cases}$$

Find the unique solution modulo 77.

Problem 6.

Solve the system

$$\begin{cases} x \equiv 2 \pmod{5}, \\ x \equiv 4 \pmod{9}, \\ x \equiv 3 \pmod{7}. \end{cases}$$

Find the smallest non-negative solution.

Problem 7.

Find the smallest positive integer x satisfying

$$\begin{cases} x \equiv 1 \pmod{8}, \\ x \equiv 4 \pmod{9}, \\ x \equiv 6 \pmod{11}. \end{cases}$$

Give your answer modulo

$$8 \cdot 9 \cdot 11.$$

Problem 8.

Find the smallest positive integer n such that

$$\begin{cases} n \equiv 2 \pmod{5}, \\ n \equiv 3 \pmod{7}, \\ n \equiv 4 \pmod{9}, \\ n \equiv 0 \pmod{11}. \end{cases}$$

Problem 9.

Find all integers x satisfying

$$\begin{cases} x^2 \equiv 1 \pmod{8}, \\ x \equiv 2 \pmod{5}, \\ x \equiv 3 \pmod{7}. \end{cases}$$

Give all solutions modulo

$$8 \cdot 5 \cdot 7.$$

Problem 10. Challenge

Find the smallest positive integer x satisfying

$$\boxed{\begin{aligned} x^2 &\equiv 1 \pmod{16}, \\ x &\equiv 3 \pmod{11}, \\ x &\equiv 5 \pmod{13}, \\ x &\equiv 7 \pmod{17}. \end{aligned}}$$

Your solution should first determine all possible values of x modulo 16, and then use the Chinese Remainder Theorem to find the smallest positive solution.

Remark (Finding Modular Inverses).

To find the inverse of a modulo m, we seek an integer x such that

$$ax \equiv 1 \pmod{m}.$$

Such an inverse exists if and only if $\gcd(a, m) = 1$.

By Bézout's identity, if $\gcd(a, m) = 1$, we can find integers u, v such that

$$au + mv = 1.$$

Reducing modulo m gives

$$au \equiv 1 \pmod{m},$$

so

$$\boxed{a^{-1} \equiv u \pmod{m}}.$$

The integers u, v can be found using the Extended Euclidean Algorithm.

Example. Find $7^{-1} \pmod{26}$.

Using the Euclidean algorithm,

$$26 = 3 \cdot 7 + 5,$$
$$7 = 1 \cdot 5 + 2,$$
$$5 = 2 \cdot 2 + 1.$$

Now work backwards:

$$1 = 5 - 2 \cdot 2$$
$$= 5 - 2(7 - 5)$$
$$= 3 \cdot 5 - 2 \cdot 7$$
$$= 3(26 - 3 \cdot 7) - 2 \cdot 7$$
$$= 3 \cdot 26 - 11 \cdot 7.$$

Thus

$$(-11) \cdot 7 + 3 \cdot 26 = 1.$$

Reducing modulo 26 gives

$$-11 \cdot 7 \equiv 1 \pmod{26}.$$

Therefore

$$7^{-1} \equiv -11 \equiv \boxed{15} \pmod{26}.$$

Explicit Construction of the CRT Solution

Suppose

$$\begin{cases} x \equiv a_1 \pmod{m_1}, \\ x \equiv a_2 \pmod{m_2}, \\ \vdots \\ x \equiv a_k \pmod{m_k}, \end{cases}$$

where $m_1, \ldots, m_k$ are pairwise coprime.

Set

$$M = m_1 m_2 \cdots m_k, \qquad M_i = \frac{M}{m_i}.$$

Since $\gcd(M_i, m_i) = 1$, the inverse of M_i modulo m_i exists. Let

$$y_i \equiv M_i^{-1} \pmod{m_i}.$$

Then an explicit solution is

$$\boxed{x \equiv \sum_{i=1}^{k} a_i M_i y_i \pmod{M}.}$$

Indeed,

$$M_i y_i \equiv 1 \pmod{m_i},$$

while $M_i y_i \equiv 0 \pmod{m_j}$ for $j \neq i$. Thus, modulo m_i, all terms vanish except $a_i M_i y_i$, giving

$$x \equiv a_i \pmod{m_i}.$$

Example. Solve

$$\begin{cases} x \equiv 2 \pmod{5}, \\ x \equiv 3 \pmod{7}, \\ x \equiv 4 \pmod{9}. \end{cases}$$

Here

$$M = 315, \qquad M_1 = 63, \quad M_2 = 45, \quad M_3 = 35,$$

and

$$63^{-1} \equiv 2 \pmod{5}, \qquad 45^{-1} \equiv 5 \pmod{7}, \qquad 35^{-1} \equiv 8 \pmod{9}.$$

Therefore

$$x \equiv 2(63)(2) + 3(45)(5) + 4(35)(8) \equiv 157 \pmod{315}.$$

Hence

$$\boxed{x \equiv 157 \pmod{315}}.$$